

Linear Models & Interpretation

Foundation to Challenge

Practice should make the test feel easier, not merely familiar.

Work without the answer key first. Check answers through the SAT Practice Library after completing the set.

FOUNDATION

- 1** A plant height h , in cm, is modeled by $h = 5f + 12$, where f is grams of fertilizer. I. How much does one additional gram change h ? II. How many grams increase h by exactly 1 cm?

- 2** A used laptop value V from 2020 onward is modeled by $V = 1800 - 120t$, where t is years since 2020. What do 1800 and 120 represent?
- A. Initial value \$1800; decreases \$120 per year
 - B. Final value \$1800; increases \$120 per year
 - C. Average value \$1800; decreases \$120 total
 - D. Initial value \$120; decreases \$1800 per year

- 3** The water depth $h = 42 - 1.5d$ models depth after d days. How does the depth change each day?
- A. decreases 1.5
 - B. increases 1.5
 - C. decreases 42
 - D. increases 42

- 4** A bakery has $b = 260 - 22h$ loaves h hours after opening. What does 22 represent?
- A. 22 loaves remain at closing
 - B. 22 loaves are sold per hour
 - C. 22 loaves are baked per hour
 - D. the bakery closes after 22 hours

5 A utility bill is $C(e) = 0.18e + 24$. What does 24 represent?

- A. fixed monthly fee
- B. cost per kWh
- C. cost of 24 kWh
- D. maximum bill

6 A bicycle speed is $s = 18 + 3t$, where t is minutes after acceleration begins. What is s after 5 minutes?

- A. 15
- B. 18
- C. 33
- D. 45

7 A tutorial site models members by $m = 350 + 80n$, where n is the number of videos. Which statement is true?

- A. 80 members without videos
- B. 350 new members per video
- C. 350 members when $n = 0$
- D. 80 videos were initially available

STANDARD

8 A drink recipe uses $s = 14 - 1.5h$ teaspoons of sugar for h teaspoons of honey. What does 1.5 represent?

- A. 1.5 more sugar per honey teaspoon
- B. 1.5 fewer sugar teaspoons per honey teaspoon
- C. 1.5 honey teaspoons per sugar teaspoon
- D. 14 fewer sugar teaspoons

FOUNDATION

9 The temperature $T = 72 - 4m$ models a drink m minutes after cooling starts. What is the total temperature decrease after 6 minutes?

STANDARD

- 10** An auction price is $p = 1200 - 15t$, where t is seconds remaining. What does 1200 represent?
- A. starting price at the beginning of timing
 - B. price when $t = 0$
 - C. price decrease per second
 - D. number of seconds
- 11** Pollutant mass $y = 150 - 12x$ remains x hours after treatment begins. What does the x -intercept represent?
- A. initial pollutant amount
 - B. removal rate
 - C. amount after 2 hours
 - D. time until modeled amount reaches zero
- 12** A currency exchange uses $y = 0.92x - 2.5$, where x is dollars and y is euros. What does 2.5 represent?
- A. a fixed 2.5-euro fee
 - B. a 2.5-dollar exchange rate
 - C. 2.5 euros per dollar
 - D. 2.5 dollars per euro

FOUNDATION

- 13** Room temperature $T = 61 + 6h$ degrees F after h hours. What is the increase every 30 minutes?

- 14 A container has $f(x) = 30 - 5x$ ounces of juice after x servings are made. What is the y -intercept?
- A. 30 ounces initially
 - B. 5 ounces initially
 - C. 30 ounces used per serving
 - D. 5 servings total

ADVANCED

- 15 The equation $3y - x = 18$ relates gallons y to the number x of turtles. Which statement is true?
- A. one turtle needs 3 gallons
 - B. one additional turtle increases y by $\frac{1}{3}$ gallon
 - C. one gallon supports 3 turtles
 - D. one turtle decreases water by $\frac{1}{3}$ gallon

FOUNDATION

- 16 Mailing cost $C = 2 + 3x$ dollars for x pounds. A \$15 increase in cost corresponds to how many additional pounds?

- 17 A plumber charges a one-time fee plus an hourly rate. In a graph of total charge versus hours, what does the slope represent?
- A. hourly rate
 - B. one-time fee
 - C. maximum charge
 - D. total charge

FINISHED?

Check your answers online

aryalanup.name/np/sat-linear-models.html

Do not stop at “wrong.” Identify why it was wrong: concept, algebra, arithmetic, interpretation, calculator choice, or timing.

I do not correct the answer first. I correct the reason.

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Scan to check answers